

Process Control

Final Lab Report
Unit Operations Lab 12
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1.Executive Summary (10 pts)

The objective of this experiment was to evaluate the dynamic behavior of a liquid level process and determine how manual and feedback control strategies influence system stability, disturbance rejection, and steady-state accuracy. The Armfield PCT50 level control unit was used to study open-loop (manual) control as well as closed-loop proportional (P) and proportional- integral (P+I) control of water level in a tank.

In open-loop operation, a pump speed of approximately 47% maintained the water level near 100 mm under constant outflow conditions. When an additional outflow disturbance was introduced, the level decreased until the pump speed was set manually increased to restore balance. Filling tests showed rates of approximately 70.5 mm/min at 100% pump speed and 41.5 mm/min at 75%, while draining remained consistent at about 35-36 mm/min, indicating repeatable system behavior.

Under proportional-only control, decreasing the proportional band reduced steady-state offset but increased variability. Large proportional bands produced stable but significant offset, while small bands reduced offset at the cost of oscillatory behavior. Adding integral action reduced steady-state error further and improved steady state further and improved setpoint tracking. However, shorter integral times increased fluctuations, showing the tradeoff between accuracy and stability.

Overall, the results confirm expected control theory behavior for integrating level process. Manual control proved sensitive to disturbances, P-only control reduced but did not eliminate offset, and properly tuned P+I control provided the best balance between stability and accuracy.

2. Introduction (10 pts)

In this experiment, the PCT50 Level Control Process was used to study introductory process control theory through observing how the water level in a process vessel changes in response to variations in flow. For this lab, water was held inside of a reservoir and pumped into a sump tank while simultaneously draining water through either a mechanical valve, solenoid valve, or both. These valves also had the option of different orifice diameter fittings, which affected outlet flow. Level inside of the sump tank is monitored digitally and physically, and the water inside the tank remains constant when the pump speed matches the draining speed. However, both manual and digital disturbances caused the level to either fall or rise, which emphasizes the need for process controls for maintaining a constant tank level.

Responses to falling or rising water levels included manually or automatically changing the pump speed. In manual control, one needed to constantly adjust the pump speed while observing the level. When proportional control was applied, the control system automatically adjusted the pump speed by measuring the difference between the set point and offset. In this case, adding increasing integral control to the system would decrease the offset, but also increase the variability and error, which stresses the importance of process optimization in tuning level control systems.

The key objective of this experiment is to study the differences between manual level control, proportional control, and integral control within a single control loop with water as the working fluid. Data on water level and pump speed were collected over time for various disturbances, water level setpoints, solenoid valve openings and proportional band settings. This experimental data was used to evaluate the overall system efficiency, error, and adaptability. Through this analysis, the lab demonstrated fundamental principles of process control and stressed the importance of tuning controls to ensure stable operation in chemical engineering process- even those as simple as water flow.

3. Theory (10 pts)

Process Control – Level control (Armfield PCT50)

The purpose of this experiment is to study liquid level control using the Armfield PCT50 level control unit. The system demonstrates both open loops and closed loop (manual) and closed loop (automatic PID) control. The controlled variable is the water level in the process vessel, while the manipulated variable is the pump speed which determines the inlet flow rate. The behavior of the liquid level over time can be explained using a mass balance on the tank and basic feedback control principles, which describe how changes in inflow and outflow affect the level and how the controller responds. [1]

The Armfield PCT50 level-control trainer provides a controlled process vessel with instrumentation and a controller interface for adjusting proportional band and integral time during level-control experiments. [2]

Level Mass balance

Since the tank has a constant area, the volume can be written as:

Substituting into the mass balance:

Equation (1) shows that: If flow in is greater than flow out, the level rises, and If flow out is greater than flow in, the level drops, lastly If flow in equals flow out, the level stays constant. [1]

The water level in the tank is measured using a pressure sensor located at the bottom of the vessel. The sensor measures the hydrostatic pressure created by the water column above it. This relationship is given by:

Since pressure increases as the water level increases, the controller can determine the height of the water by measuring the pressure at the bottom of the tank.

Proportional (P) control

In closed-loop control, the controller continuously compares the measured water level to the desired setpoint. The difference between them is called the error:

For proportional control, the controller output is directly proportional to this error:

This means that the larger the error, the larger the change in pump speed.

In the PCT50 system, the controller is set using Proportional Band (PB) instead of gain. The relationship between them is:

In proportional-only control, a nonzero steady-state error (offset) can remain because the controller needs some sustained error to hold a nonzero output. In the proportional band form used here, the proportional band is inversely related to gain (), so smaller PB corresponds to stronger proportional action. [3]

A large proportional band gives a slower response and usually results in a larger steady state offset. A smaller proportional band makes the system respond faster, but if it is set too low, the system may start to oscillate or become unstable.

To fix the steady-state error that happens with P-only control, integral action is added to the controller:

The integral part works by adding up the error over time. If the water level stays below the setpoint, the controller keeps increasing the pump speed until the error is removed. Because of this, PI control can eliminate the steady state offset. [4]

However, if the integral time is too small, the controller reacts too quickly and too strongly. This can cause the level to overshoot and start oscillating, which makes the system unstable.

Table 1 – Variables and Units

Variable	Units
Water Level (<i>h</i>)	<i>mm</i>
Time (<i>t</i>)	<i>s</i>

<i>Pump Speed</i>	<i>%</i>
<i>Volumetric Flow Rate (Q)</i>	<i>m³/s</i>
<i>Tank Area (A)</i>	<i>m²</i>
<i>Pressure (P)</i>	<i>Pa</i>
<i>Density (ρ)</i>	<i>kg/m³</i>
<i>Gravity (g)</i>	<i>m/s²</i>
<i>Proportional Band (PB)</i>	<i>%</i>
<i>Integral Time (τ_i)</i>	<i>s</i>

4. Results (15 pts)

Experiment 1

Experiment A examined manual (open-loop) level control on the Armfield PCT50 by adjusting pump speed to maintain water level, then applying disturbances by toggling the solenoid valve. The experiment also measured filling and draining rates by recording level vs. time during controlled fill and drain periods.

At a constant 47% pump speed, the water level stayed in a relatively narrow band around roughly ~98–101 mm, indicating that a steady level can be maintained in open-loop operation before any disturbances are introduced. This trial establishes the baseline pump setting needed to hold the level without disturbances. The short, sharp dips (near ~1.7–2 min) appear as brief outliers, but the level returns to the previous range afterward, so overall stability remains acceptable. These results can be seen in Figure 1.

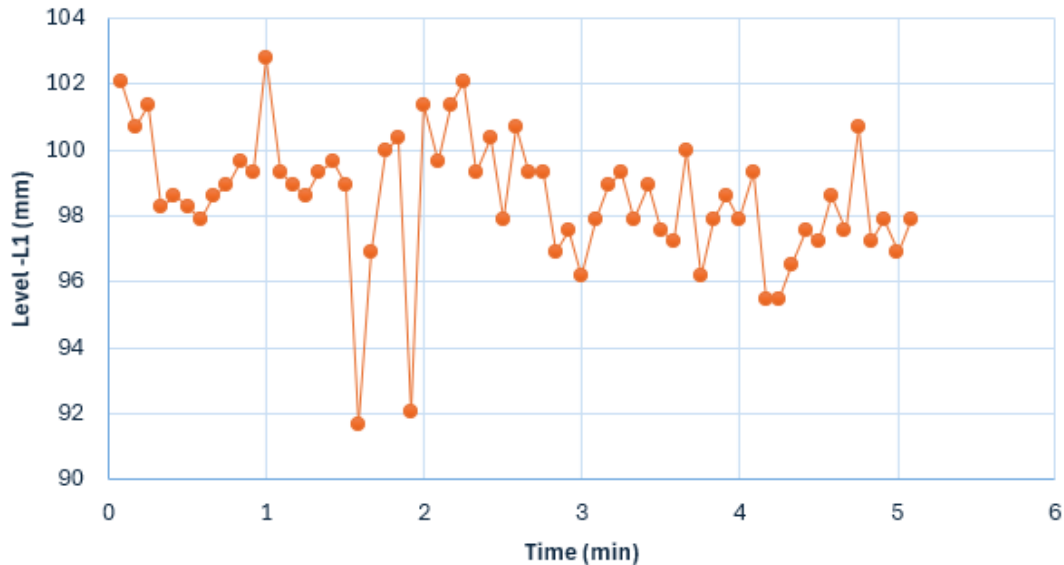


Figure 1: Initial Level Control (47% pump speed)

When the solenoid was opened, the level steadily decreased at the first pump setting (~48%), indicating that the outflow disturbance exceeded the inflow at that setting. This plot is meant to show the level response when an extra outflow path is added and how much the pump must be increased to compensate. Increasing the pump speed to about 51% reversed the trend and the level recovered upward, then settled near a higher level, showing that increasing inflow can offset the added outflow disturbance. These results can be seen in Figure 2.

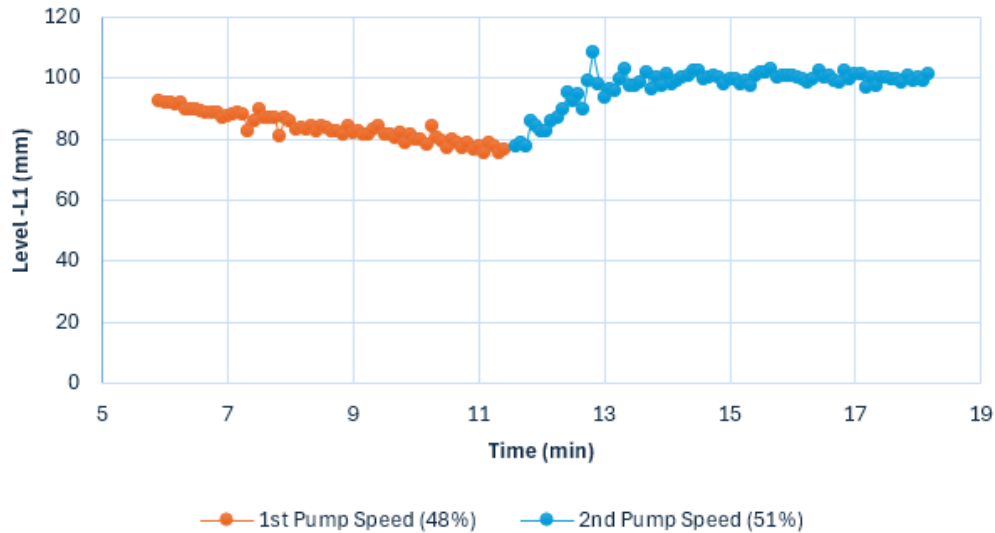


Figure 2: Solenoid On at 1st and 2nd Pump Speed (Disturbance Added)

With the solenoid closed, the level initially increased at the second pump setting (~51%), consistent with reduced outflow compared to the solenoid-open case. This plot is meant to show the corrective pump adjustment after removing the disturbance, followed by the draining behavior once the pump is shut off. Reducing the pump to about 45% caused the level to decrease back toward ~100 mm. Afterward, turning the pump off produced a rapid, smooth draining segment toward near-zero level. These results can be seen in Figure 3.

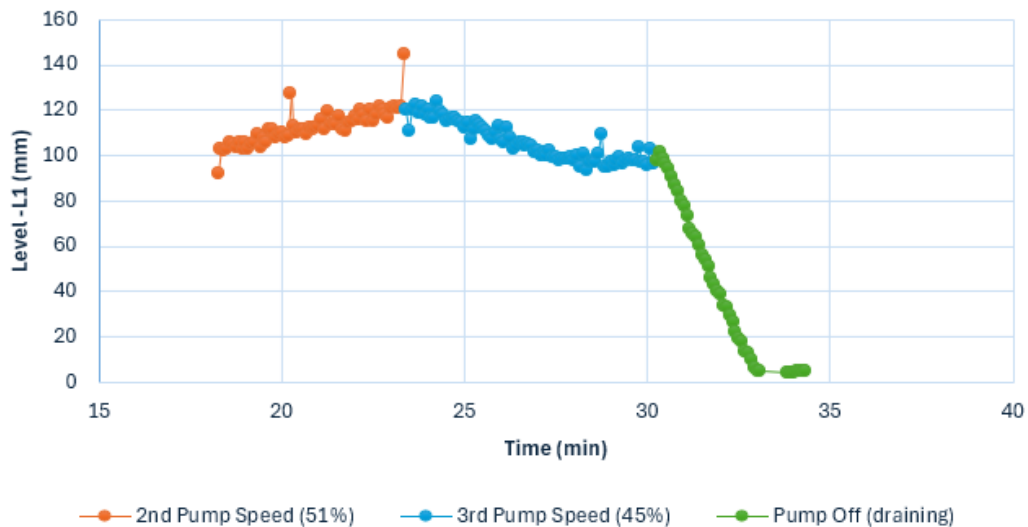


Figure 3: Solenoid Off at 2nd and 3rd Pump Speed (+ Drain)

During filling at 100% pump speed, level increased nearly linearly with time. This plot is meant to estimate the filling rate at a high pump setting using the slope of level vs. time. The trendline slope (~70.5 mm/min) represents the filling rate under this condition, and

the high supports using the slope as a summary of the fill rate. *These results can be seen in Figure 4.*

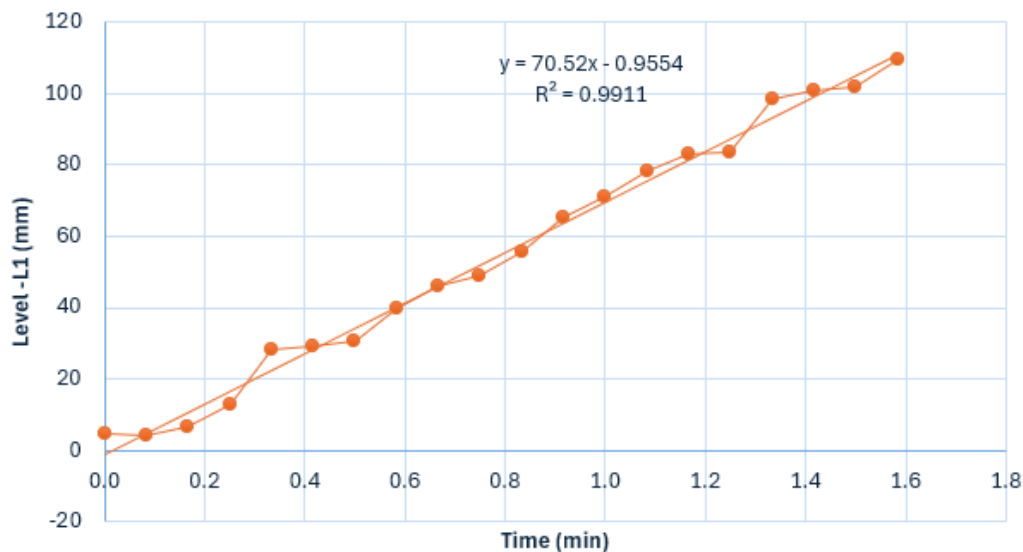


Figure 4: Vessel Filling (100% speed)

With the pump off (0%), the vessel drained at an approximately linear rate with a trendline slope of ~ 35.4 mm/min (negative on the plot, reported here as a magnitude), suggesting the draining flowrate stayed fairly constant over most of the interval. This plot is meant to estimate the natural drain rate of the system with no pumping. If two drain trials were taken, the two measured draining slopes (35.427 and 36.641 mm/min) are close (about a 3% difference), suggesting the draining flowrate was essentially the same between trials. *These results can be seen in Figure 5.*

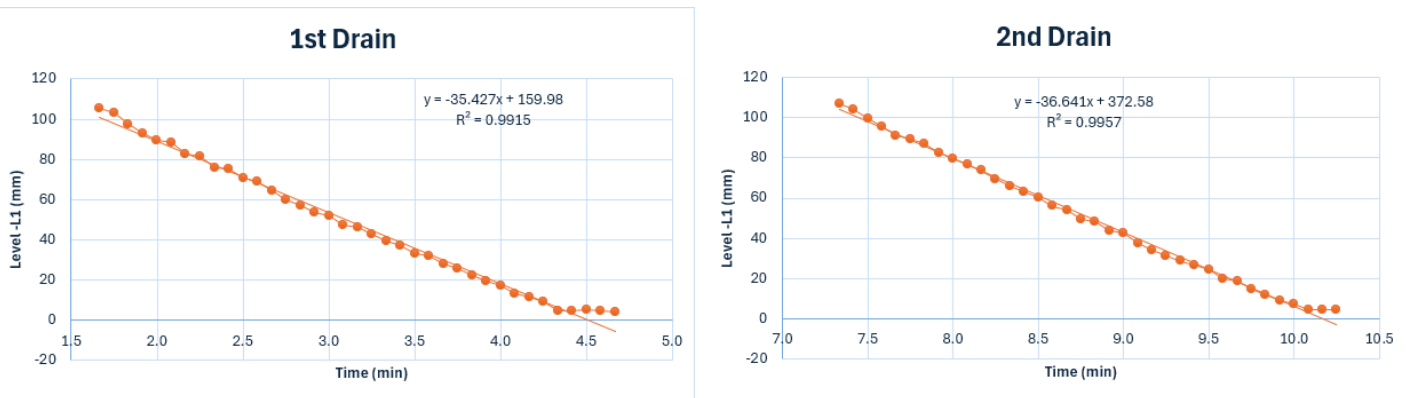


Figure 5: Vessel Draining (0% speed)

At 75% pump speed, the level increased nearly linearly with a trendline slope of ~ 41.5 mm/min, which represents the filling rate for this setting. This plot is meant to compare how pump speed affects the filling rate using the same level vs. time approach as Figure 4. Comparing the filling slopes shows the filling flowrate did not stay the same between pump speeds: the 100% setting filled substantially faster than 75%, as expected. These results can be seen in Figure 6.

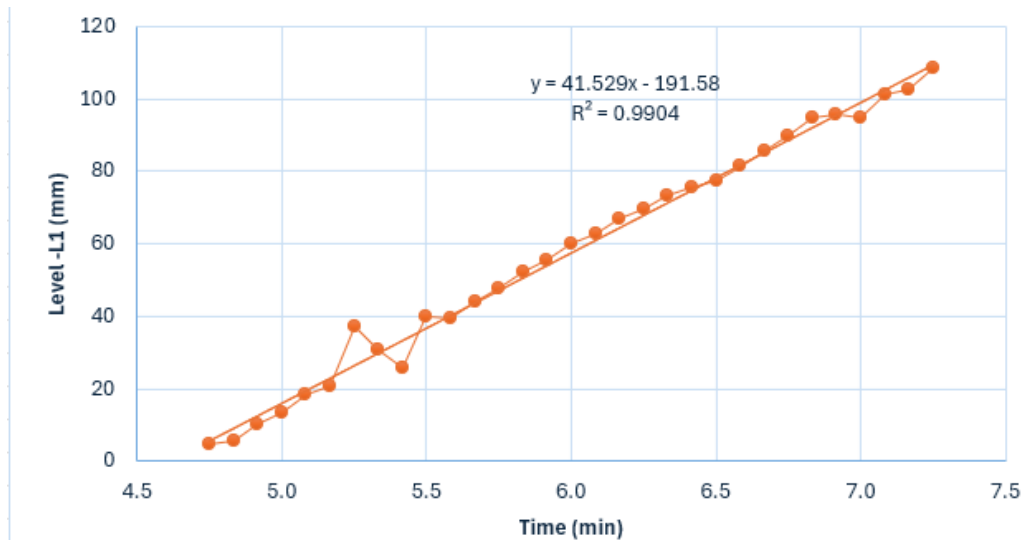


Figure 6: Vessel Filling (75% speed)

Overall, Experiment A showed that the vessel level can be maintained in open-loop operation with an appropriate pump setting, but it shifts predictably when outflow disturbances are added or removed. The fill and drain trials also gave consistent, near-linear rates that can be used as baseline flow behavior for later closed-loop control.

Experiment 2

Experiment B evaluated P-only closed-loop level control by varying the Proportional Band (P) and observing how the process responds under (1) normal operation at a 75 mm setpoint, (2) an outflow disturbance (solenoid opened), and (3) a setpoint change to 100 mm. For each P value, the offset (steady difference between measured level and setpoint) and the stability (amount of fluctuation/oscillation) were recorded and compared.

At $P = 200\%$, the level increases slowly and then settles with small fluctuations, but it remains noticeably below the 75 mm setpoint (large steady offset). The response is smooth overall, indicating conservative control action and strong damping. These results can be seen in Figure 7.

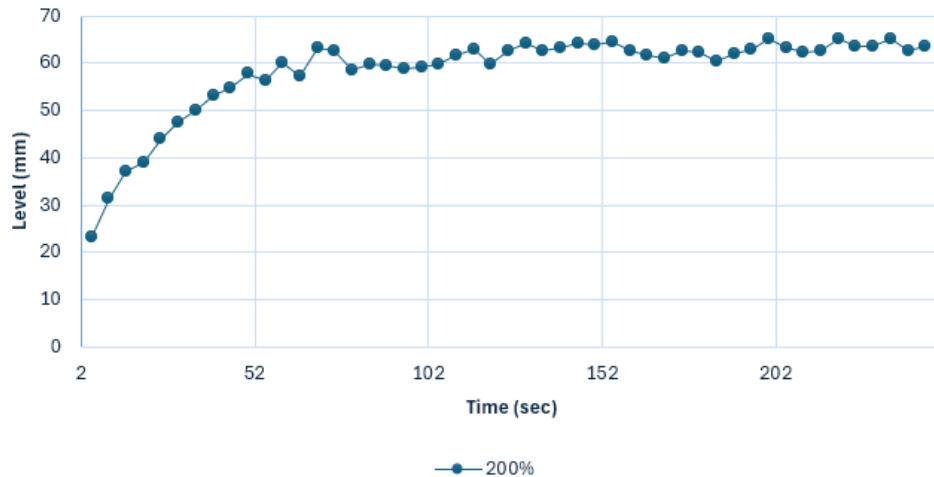


Figure 7: 200% Proportional Band at 75 mm setpoint

With the solenoid opened at $P = 200\%$, the level stays relatively steady with only small variations, but it continues to show a clear offset from the setpoint. Compared to the normal case, the disturbance does not create large oscillations, but the controller does not eliminate steady-state error under P-only control. These results can be seen in Figure 8.

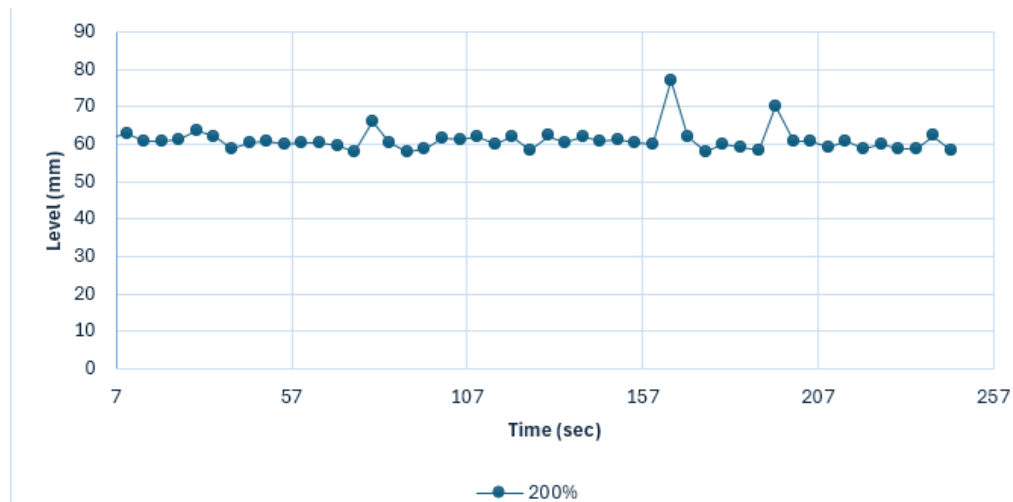


Figure 8: 200% Proportional Band with open valve

After the setpoint change to 100 mm at $P = 200\%$, the level rises and then stabilizes with limited variation, but it still does not settle exactly at the new setpoint (offset remains). Overall, the response is stable and not oscillatory, but the steady-state error persists. These results can be seen in Figure 9.

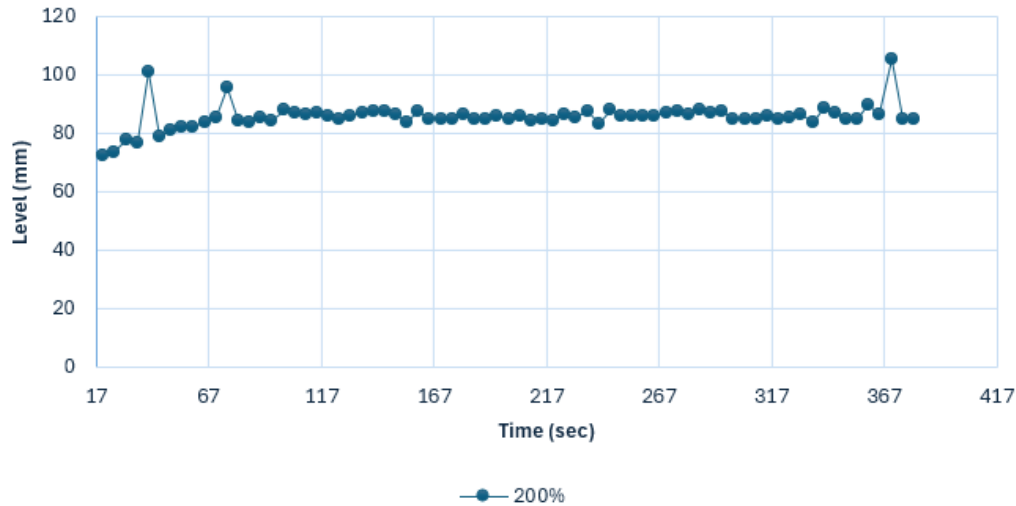


Figure 9: 200% Proportional Band at 100 mm setpoint

Across the other P-band values (100% → 2%), the main trend is that lowering the proportional band generally reduced the average offset in all three cases (normal, solenoid disturbance, and 100 mm setpoint change). This matches what you'd expect from P-only control, where more aggressive proportional action pushes the level closer to the setpoint, so the steady-state error tends to shrink. In our data, the higher P-band settings (200% and 100%) consistently showed the largest offsets, while the smaller P-band settings clustered at lower offsets overall. These trends can be seen in Figures 13–18 in the data tabulations section, and the offsets are summarized numerically in Table 2.

Table 2: Offset and stability for variable proportional band

Proportional Band	Normal		disturbance		Setpoint Change	
Property	offset (mm)	stability note	offset (mm)	stability note	offset (mm)	stability note
200%	16.7	very stable	14.0	very stable	15.0	very stable
100%	6.6	stable	7.3	stable	10.4	stable
50%	3.8	stable	4.1	stable	6.2	stable
20%	3.0	somewhat stable	2.0	somewhat stable	3.3	somewhat stable
10%	2.5	somewhat stable	2.4	somewhat stable	2.9	somewhat unstable
5%	2.1	somewhat unstable	1.5	somewhat unstable	4.0	erratic
2%	1.6	erratic	3.2	erratic	2.5	erratic

The stability trend is less “perfectly monotonic,” but the pattern is still obvious. as P-band decreases, the response becomes more sensitive, which can increase visible fluctuations in the level traces. In the mid-range settings (around 50%–20%), the level traces generally look closer to the setpoint than the higher P-band cases, while still staying fairly controlled. At the smallest settings (like 5% and 2%), the offset is small, but the level signal shows more noticeable variation in some cases, especially when disturbances

or setpoint changes are applied. This tradeoff—smaller offset vs. more variability—is the main reason we used both the plots (Figures 13–18) and the offset/stability table to choose a P value for the next experiment.

Because 2% proportional band gave the smallest overall offsets while still keeping the level stable enough to run the next test, we selected 2% P-band as the proportional setting to carry into Experiment 3 (P+I control).

Experiment 3

In Experiment 3, the controller was run in closed-loop P + I mode using the selected proportional band from Experiment B while the Integral Time (I) was varied (100, 50, 25, 10 s). The goal was to see how changing I affects the level response, offset from the setpoint, and stability under normal operation, an added outflow disturbance (solenoid open), and a setpoint change.

For the 75 mm setpoint with the solenoid off, the level stays in the general 70–85 mm range for all I settings, with visible fluctuations around the setpoint rather than a perfectly flat line. The traces also show that the response behavior changes with I (some I values drift more and/or fluctuate more than others), but none of the runs shown here look like runaway instability within the plotted time window. These results can be seen in Figure 10.

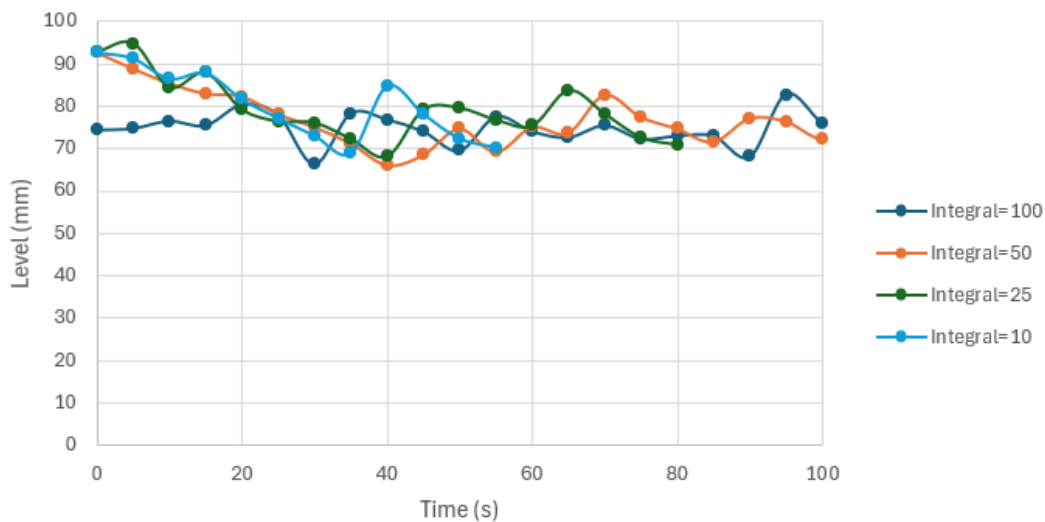


Figure 10: solenoid off at 75 mm setpoint

When the solenoid is turned on at the 75 mm setpoint, the level still remains near the setpoint overall, but the curves show larger short-term swings and occasional spikes compared to the solenoid-off case, consistent with the added outflow disturbance. Across the different I values, the response remains bounded in the plotted interval, but it is clearly “noisier” than normal operation. These results can be seen in Figure 11.

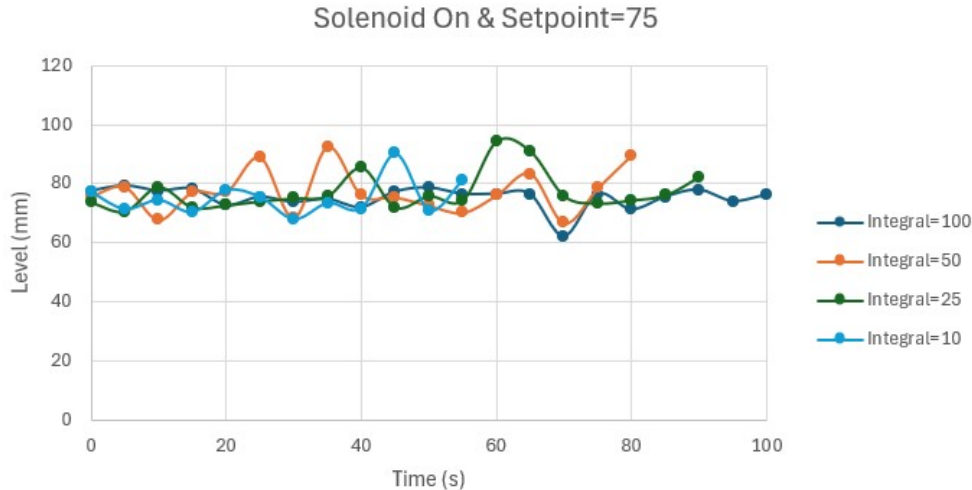


Figure 11: solenoid on at 75 mm setpoint

For the 100 mm setpoint with the solenoid off, the level generally stays around ~95–110 mm across the tested I values, again with noticeable oscillations/spikes rather than a perfectly steady line. The curves suggest the controller can hold the level near the higher setpoint, but the response is still not perfectly smooth within the 0–100 s window shown. These results can be seen in Figure 18.

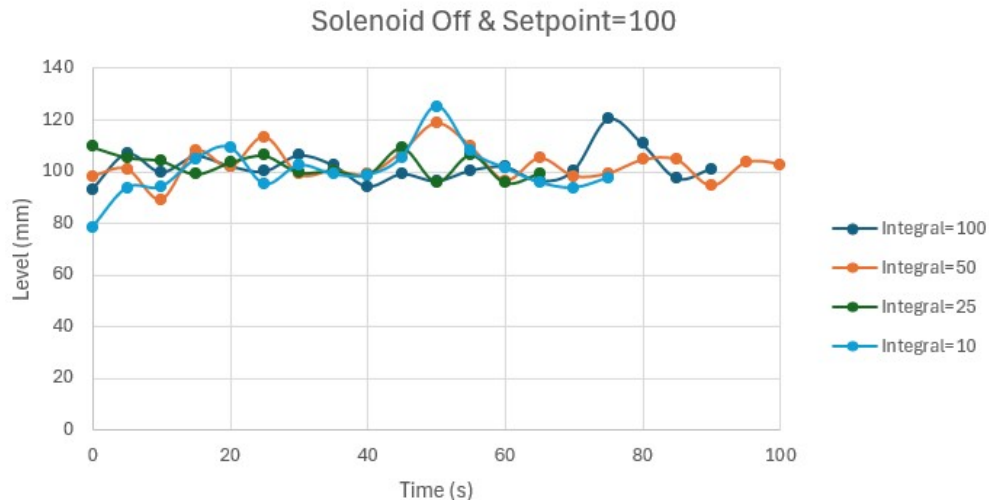


Figure 12: solenoid off at 100 mm setpoint

The average levels and offsets computed from the data are summarized in Table 3. In our trials, the smallest offsets at the 75 mm setpoint occurred at the larger integral times ($I = 100$ s gives ~2.08 mm with solenoid off and ~2.53 mm with solenoid on), while the largest offset occurred at $I = 10$ s for the solenoid-off case (~7.99 mm). For the 100 mm setpoint (solenoid off), offsets stayed around ~4–6 mm across the I values tested, with the smallest value at $I = 25$ s (~4.10 mm). This trend is reasonable because changing I mainly

changes how aggressively the controller “pushes” to correct error, which can shift the average level and how tight the response looks, as summarized in Table 3.

Table 3: Average Level and Offset (P+I)

Integral Time	Setpoint	Solenoid	Level	Offset
100	75	Off	75	2.08
	75	On	75.21	2.53
	100	Off	101.76	4.35
50	75	Off	76.02	3.48
	75	On	77.38	5.54
	100	Off	102.04	4.33
25	75	Off	79.33	6.21
	75	On	76.96	3.96
	100	Off	102.37	4.10
10	75	Off	80.38	7.99
	75	On	75.14	4.37
	100	Off	101.63	5.93

Overall, the results show how the level-control system responds to disturbances and controller settings. In open-loop operation, holding a constant pump speed maintained the level only within a limited band, and disturbances from the solenoid required manual pump adjustments to recover the level. In closed-loop control, changing the proportional band and integral time shifted the tradeoff between offset and the “tightness” of the response, with some settings producing smaller offsets but more variability.

5. Discussion (15 pts)

Experiment 1

In Experiment A, the system was operated manually, so the level depended completely on how well the pump speed matched the outflow. The results followed what would be expected for a basic tank system. When inflow and outflow were balanced, the level stayed steady. When they were not balanced, the level either increased or decreased until it was corrected.

At around 47% pump speed, the water level stayed between 98 and 101 mm. It did not drift up or down over time, which shows that the inflow at the setting was close to the outflow through the drain. There were small fluctuations in the graph, but they were minor and the level stayed within the same range. This was likely due to the slight pump variation or measurement noise rather than actual instability. Overall, that pump speed was enough to hold the level steady under normal conditions.

When the solenoid valve was opened, the level started to drop at the original pump setting of about 48%. This makes sense because opening the solenoid valve increased the total outflow. Since the pump speed was not changed at first, more water was leaving than entering, so the level decreased. Increased the pump speed to 51% brought the level back up and allowed it to stabilize again. The adjustment needed was small but noticeable, which shows how even small disturbance can shift the level when operating in open-loop mode.

After closing the solenoid valve while still running at the higher pumps speed, the level increased because the inflow was now greater than the outflow. Reducing the pump speed to around 45% brought the level back down to the original value. The response in both directions was smooth and gradual, which is typical for a tank level process. The system reacted in a predictable way every time the flow balance changed.

The filling and draining tests also matched expectations. At 100% pump speed, the level increased almost linearly with time, with a slope of about 70.5 mm per minute. At 75% pump speed, the slope was lower which was around 41.55 mm per minute. This shows that higher pump speed clearly increases the filling rate. When the pump was turned off, the vessel drained at about 35-36 mm per minute, and the two drain trials were very close to each other. The small difference suggests that the drain's behavior was consistent and repeatable.

One thing that became clear in this experiment is that maintaining the level manually is not very practical. Even though it is possible to reach steady state, any disturbance causes the level to shift, and it takes time to correct it. The operator has to continuously watch the system and adjust. This shows the limitation of open-loop control and explains why closed-loop control is needed in real processes.

Overall, the trends seen in Experiment 1 were consistent with how a liquid level system should behave. The results were repeatable, the responses to disturbances made sense, and the filling and draining rates were consistent. This experiment provides a good baseline before moving on the closed-loop control experiments.

Experiment 2

Experiment 2 evaluated proportional-only control by adjusting the proportional band and observing how the level responded under normal operation, during an outflow disturbance, and after a setpoint change. The overall trends followed what would be expected from P-only control of a level process.

At the largest proportional band setting of 200%, the response was very smooth and stable, but the level settled noticeably below 75 mm setpoint. The steady state offset was very large, around 16.7 mm under normal operation and remained significant during both the disturbance and setpoint change cases. Even though the system was very stable with minimal oscillation, the controller was too conservative to drive the level close to the setpoint. This behavior is consistent with proportional-only control, where steady-state error remains because the controller's output depends directly on error. If the error becomes small, the control action also becomes small, preventing the system from reaching the target [5].

As the proportional band was reduced from 200% down to 50% and 20%, the steady state offset consistently decreased. For example, at 100% the offset dropped to around 6-7mm, and by 20% it was closer to 2-3 mm depending on the case. This shows that stronger proportional action pushes the level closer to the set point. The system became more responsive, and the average deviation from the target was smaller. This matches the expected trend that reducing proportional band (increasing controller aggressiveness) reduces steady-state error in P-only control.

However, the improvement in offset came with a tradeoff. At lower proportional band values such as 10%, 5%, and especially 2%, the level traces showed more visible fluctuations. While the offset was smaller, the system became more sensitive to disturbance and measuring noise. In the data shift toward "somewhat unstable" or "erratic", particularly during disturbances and setpoint changes. This behavior is typical when proportional gain is too aggressive. The controller reacts strongly to small errors, which can amplify fluctuations and produce oscillatory behavior.

The disturbance tests with the solenoid valve further confirms these trends. At high proportional band (200%), the disturbance caused little oscillation but maintained a large offset. At lower proportional bands, the controller responded more strongly to the disturbance, reducing the offset but increasing variability in the level signal. The same pattern appeared during the step change to a 100 mm setpoint. Higher proportional bands gave smooth but offset responses, while lower bands improved accuracy at the cost of stability.

Overall, the results clearly show the fundamental limitation of P-only control for an integrating process like tank level. Although proportional action improves response speed and reduces offset as the band decreases, it cannot eliminate steady-state error entirely. At the same time, making the controller too aggressive introduces instability and erratic behavior. Based on the balance between reduced offset and acceptable stability, the 2% proportional band was selected for use in the next experiment with integral control, since it provided the smallest overall offsets while remaining stable enough to proceed.

Experiment 3

Experiment 3 examined how adding integral action affected the level response while keeping the proportional band fixed at 2%. The integral time was varied between 100, 50, 25, and 10 seconds to observe how the system responds under normal operation, during a disturbance (solenoid open), and after a setpoint change. The main goal was to see how integral action influenced steady-state offset and stability.

At the 75 mm setpoint with a solenoid off, the level remained near the setpoint for all integral times tested, generally within the 70-85 mm range. Unlike proportional-only control, the average offsets were much smaller, especially at the larger integral times. For example, at $I = 100$ s, the offset was about 2.08 mm, while at $I = 50$ s it was about 3.48 mm. This shows that integral action helped reduce steady state error compared to the earlier P-only results. However, as the integral time decreased, the offset increased in some cases. At $I = 10$ s, the offset rose to about 7.99 mm for the solenoid-off case. This suggests that overly aggressive integral action can worsen performance instead of improving it.

When the solenoid valve was opened at the 75 mm setpoint, the level remained close to the target overall but showed more visible fluctuations and occasional spikes. Even though the system stayed bound and did not run away, the signal appeared noisier than under normal conditions. The offset under disturbance was still relatively small at larger integral times, around 2.53 mm at $I = 100$ s, but increased at shorter integral times. This indicates that integral action improves disturbance rejection, but too small of an integral time can make the system more sensitive.

At the 100 mm setpoint with the solenoid off the controller was able to hold the level near the new target across integral times tested. The offset was generally in the range of about 4-6 mm, with the smallest value around 4.10 mm at $I = 25$ s. The level traces showed oscillation and small spikes rather than perfectly smooth settling. This suggests that while integral action reduces steady-state error, it does not eliminate short-term variability, especially when the tuning becomes more aggressive.

Overall, Experiment 3 demonstrates the tradeoff introduced by integral control. Larger integral times provided smaller offset and smoother responses, while shorter integral times increased responsiveness but introduced more variability. None of the tested cases showed complete instability within the observed time window, but the smaller integral times clearly produced tighter yet more fluctuating behavior. These results are consistent

with expected controller behavior: Integral action helps eliminate steady-state error, but if tuned too aggressively, it can amplify oscillations and reduce stability.

Together with Experiments 1 and 2, these results confirm proper tuning requires balancing accuracy and stability. Integral control improves steady-state performance compared to proportional-only control, but the choice of integral time must be made carefully to avoid excessive variability.

6. References (10 pt)

[1] Department of Chemical Engineering. *Process Control: Level Control*. University of Illinois at Chicago, 2025.

[2] Armfield Ltd. *PCT50 Process Control Trainer (Datasheet)*. Armfield, n.d. Accessed February 16, 2026.

[3] Wikipedia contributors. "Proportional Control." Wikipedia. Last modified August 13, 2025. Accessed February 16, 2026.

[4] Wikipedia contributors. "PID Controller." Wikipedia. Last modified February 4, 2026. Accessed February 16, 2026.

[5] Seborg, Dale E., Thomas F. Edgar, Duncan A. Mellichamp, and Francis J. Doyle III. *Process Dynamics and Control*. 3rd ed. Hoboken, NJ: John Wiley & Sons, 2011.

7. Appendix I: Data Tabulation/Graphs (10 pts)

The majority of the data was collected as an Excel file directly from the lab computer connected to the PID controller. Because the raw dataset is extensive, it is not reported in full in this report. Instead, the Excel file is included as an attachment. The figures below provide the level vs. time plots for the additional proportional band settings discussed in Experiment B.

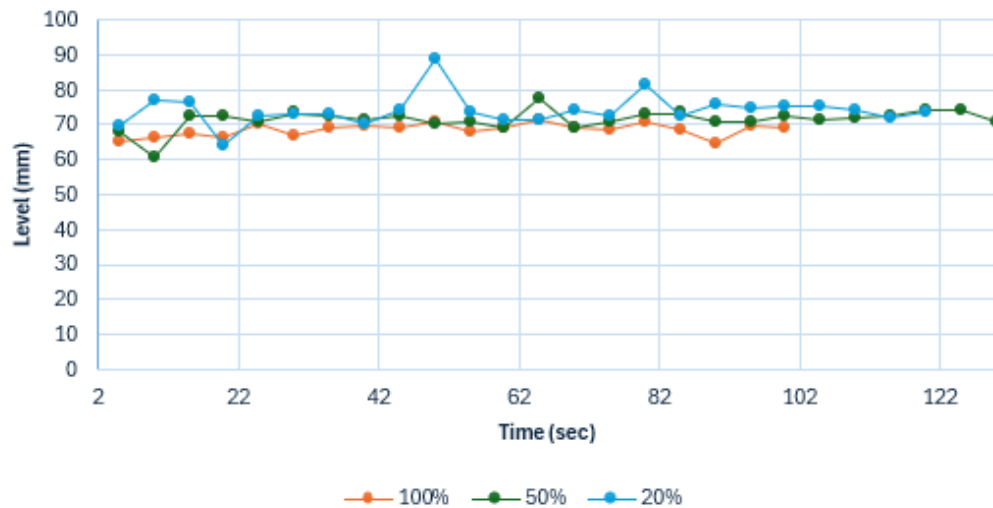


Figure 13: 100%→20% Proportional Band at 75 mm setpoint

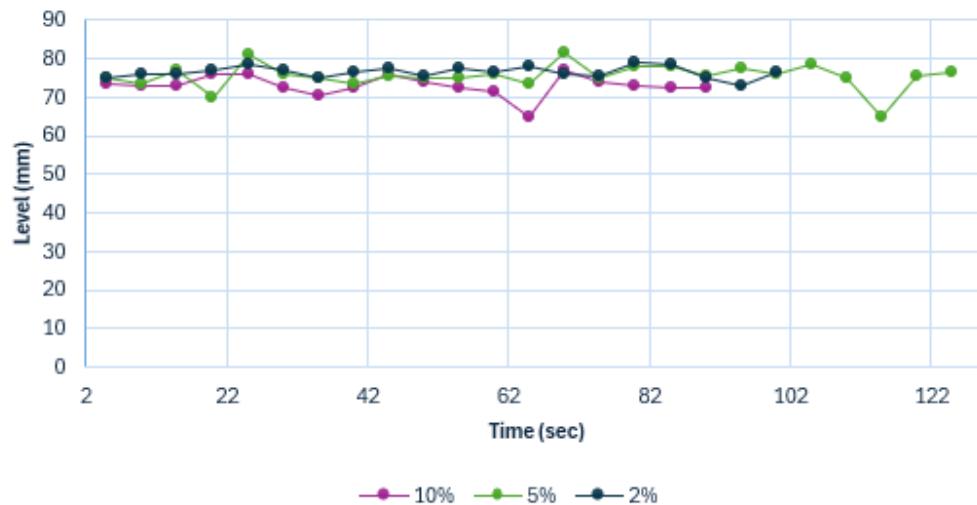


Figure 14: 10%→2% Proportional Band at 75 mm setpoint

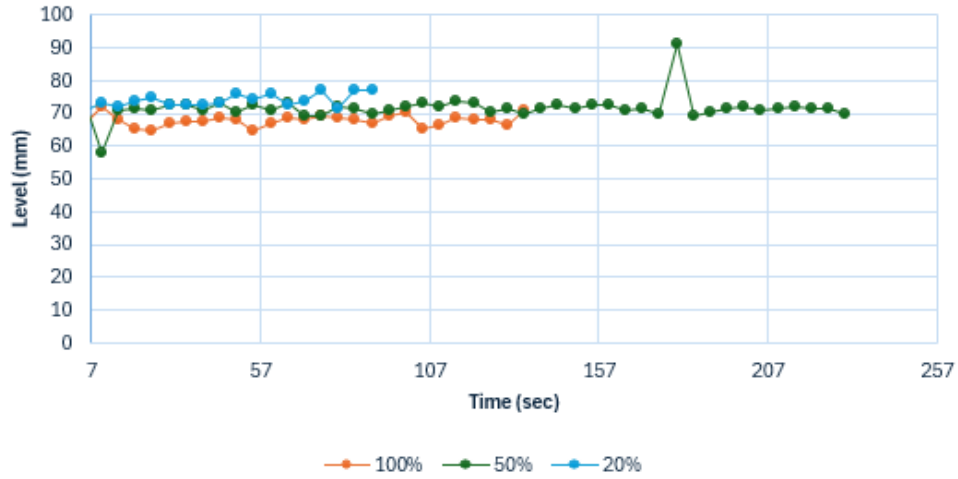


Figure 15: 100%-20% Proportional Band with open valve

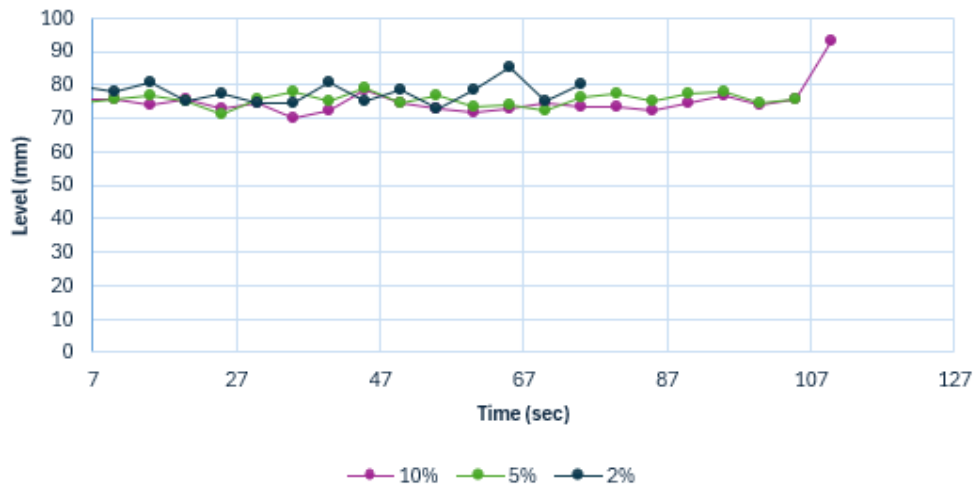


Figure 16: 10%-2% Proportional Band with open valve

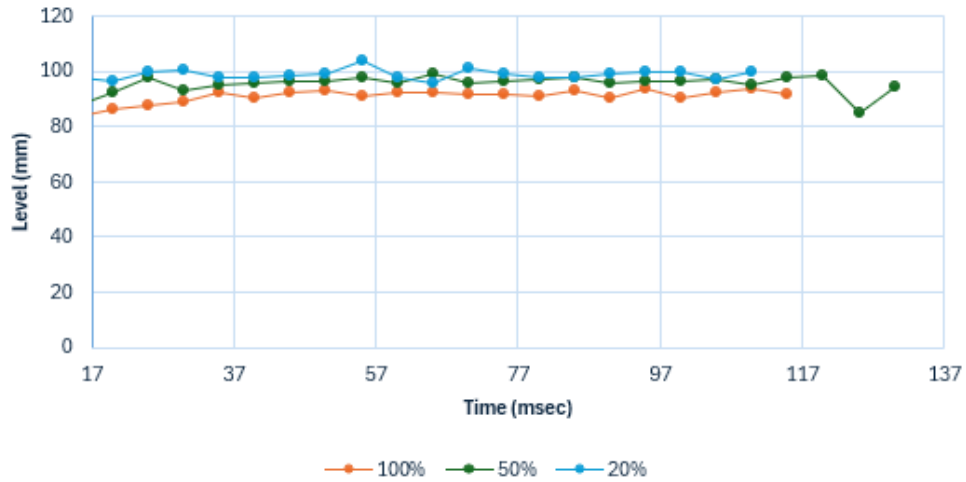


Figure 17: 100%→20% Proportional Band at 100 mm setpoint

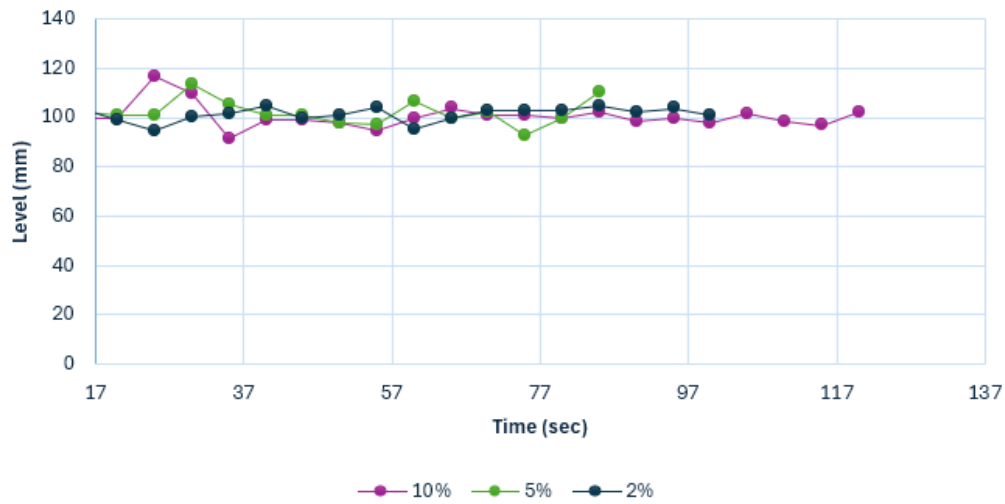


Figure 18: 10%→2% Proportional Band at 100 mm setpoint

8. Appendix II: Error Analysis (10 pts)

For this lab, a large majority of the possible areas of error came from the 1st experiment, as it required manually activating disturbances and solving them by varying pump speed. Throughout the experiment, there was difficulty with ensuring the water in the sump tank was at a constant level. There was no single speed of the pump that maintained a constant starting level of 90mm, and thus the level would drop slightly at a speed of 46 on the pump but rise slightly at a speed of 47. This uncertainty allowed there to be a constant variation in level reading from the sump tank, and thus values were noted to be around 2-3mm off for the 1st experiment. The error associated with variations of sump tank level can be easily calculated to be:

Other unavoidable errors that were noticed immediately involved the offset in level reading that the system automatically calculated. To begin the first experiment, a set point of 90mm was manually set by varying the pump speed and visually observing the tank. When at 90mm in the tank, the computer instead calculated 100mm, which is an error of about 10mm. This offset error is roughly:

Throughout Experiment 2 and 3, these offset errors persisted. However, these experiments aimed to optimize the automatic process control system through varying the Proportional Band and Integral Time to reduce the offset and reach the setpoint after a disturbance quickly and efficiently. Thus, the 11.11% error in the manual section was likely the largest consistent error throughout the experiment.

A big learning point from this lab was optimizing the process controls to a specification. If there was an error allowance of 10mm of offset in the system, there would be no need for optimization of the integral time or proportional band of the controller. However, if we aimed to have an offset allowance of ~2mm, we'd have to find the perfect combination of P and I. In this way, we are not only learning the fundamentals of process optimization but also exploring how to minimize errors in data collection.

9. Appendix III: Sample Calculations (10 pts)

The unsteady state mass balance on the tank is written as:

Since the tank has constant cross-sectional area

Substituting into the mass balance gives:

Where A is the tank area (m^2), h is the water level (m), Q_{in} is the inlet flow rate (m^3/s), and Q_{out} is the outlet flow rate (m^3/s).

When the pump is off. The mass balance simplifies to

$$A \left(\frac{dh}{dt} \right) = Q_{out}$$

Two points were selected from the draining region

$$\begin{aligned} T_1 &= 120 \text{ s}, & h_1 &= 95 \text{ mm} \\ T_2 &= 180 \text{ s}, & h_2 &= 57 \text{ mm} \end{aligned}$$

Slope calculation:

Converted to meters per second: $-6.3 \times 10^{-4} \text{ m/s}$

The negative sign shows that the level is decreasing which means the outflow is greater than inflow

At 100% pump speed:

At 75% pump speed:

The larger the positive slope at 100% confirms that increasing pump speed increases inlet flow rate

Offset is defined as:

$$\text{Offset} = \frac{Q_{in} - Q_{out}}{K}$$

Given:

$$\text{Offset} = -5.26 \text{ mm}$$

The negative value shows the steady state level is below the setpoint

Relationship between proportional band and gain

Decreasing PB increases controller gain, reducing offset but potentially causing oscillations

Stability was evaluated using oscillation amplitude:

$$\text{Amplitude} = 78.4 - 74.9 = 3.5 \text{ mm}$$

The pi controller equation is

Integral action accumulates error over time and reduces steady state offset.

$$\text{Offset} = 77.77 - 75 = -0.23 \text{ mm}$$

10. Appendix IV: Individual Team Contributions

	Time (Hours)			
	Joshua Olusoji	Omar Zaruk	Sanad Abu Hweij	Qussay Suleiman
Operator (Both Lab Days)	4	5	4	3
Pre-Lab Editing	1	1	2	2
Experimental Objective (Prelab)				1
Apparatus (prelab)	2			
Materials & Supplies (prelab)			1	
Procedure (prelab)		1		
Project Safety Evaluation (prelab)				2
Final Lab Editing	1	1		1
Executive Summary (final lab)		1		1
Introduction (Final lab)	2			
Theory (Final Lab)			4	
Results (final lab)		7		4
Discussion (final lab)				
References (final and/or prelab)	1	1		0.5
Data Tabulation/Graphs (final)		1		
Error Analysis (final lab)	2			
Sample Calculations (final Lab)			4	
Power Point Presentation	2		1.5	2
Total	15	18	16.5	16.5